That’s a great scope. Here’s a practical, end-to-end architecture plan to build a CloudFront-based CDN on AWS with Cognito, DynamoDB, S3, a video-then-transcode pipeline, a paywall, Lightsail for admin UI, IAM, CloudFormation, CloudWatch, and billing/management. I’ve kept it modular so you can start small and grow.

1) Target architecture (high-level)
- Edge/CDN
  - CloudFront distribution with S3 as origin (videos/assets) and a separate origin for dynamic API (API Gateway/Lambda) if needed.
  - Lambda@Edge or CloudFront Functions at edge to enforce paywall, user authentication, and simple access checks (using Cognito JWTs).
  - CloudFront signed URLs or signed cookies for paywalled content.
- Identity and access
  - Cognito User Pool for sign-in/sign-up.
  - Cognito Identity Pool for temporary AWS credentials if you need direct AWS access from the client (optional; often not required).
- Storage and content
  - S3 buckets for videos, thumbnails, and static assets (versioning, encryption at rest, lifecycle policies).
  - S3 bucket policies and an Origin Access Identity (OAI) to restrict direct access to videos.
- Metadata and state
  - DynamoDB tables for: Users/Subscriptions, Content catalog, Access logs, and Paywall rules.
  - Optional: RDS/Aurora if you need complex relational queries (not required for most CDN use cases).
- Video processing and AI
  - MediaConvert for video transcoding (e.g., multiple bitrates/resolutions).
  - Step Functions to orchestrate ingest → transcode → post-process (watermarking, thumbnail generation, captions).
  - SageMaker for AI functionalities (captioning, content moderation, recommendations) or for custom inference on video-related tasks.
  - Optional: Lambda or SageMaker endpoints for on-the-fly access control or personalization signals.
- Admin and management
  - Lightsail-hosted admin UI or microservices that manage content, subscriptions, and analytics.
- Orchestration and IaC
  - CloudFormation templates (modular stacks) to deploy/core resources, with parameterization per environment (dev/stage/prod).
- Observability and cost mgmt
  - CloudWatch for logs, metrics, dashboards, alarms.
  - CloudTrail for API activity.
  - AWS Budgets and Cost Explorer for billing/finOps.
- Security and compliance
  - IAM least-privilege roles, KMS keys for encryption, WAF for CloudFront, Shield for DDoS protection.
  - VPC where needed (private subnets for Lambda/Lightsail apps, VPC endpoints for private service access).

2) Data flow overview
- User journey: Web/mobile client → CloudFront
  - If user requests paywalled content, edge logic (Lambda@Edge) validates Cognito JWT or cookies.

  - On unauthenticated/unauthorized requests, redirect to Cognito-hosted UI or API-based login flow.
  - Authorized requests fetch from S3 via CloudFront with signed URLs/cookies, or via an API that confirms subscription status in DynamoDB.
- Content and video pipeline:
  - Ingested video lands in S3.
  - S3 event triggers a Step Functions workflow: prepare metadata → start MediaConvert job(s) for multiple renditions → post-process (thumbnails, captions) → store results back in S3; update DynamoDB with new video metadata.
  - Optional AI tasks run in SageMaker (captioning, auto-moderation, recommendations) and push results back to DynamoDB or a dedicated model endpoint.
- Paywall and subscription:
  - Client app queries Subscription state from DynamoDB (via API Gateway/Lambda or directly from the app with Cognito JWTs).
  - Paywall gating in the app or via edge logic enforces access to video assets.
- Admin/UI:
  - Lightsail hosts the admin UI that uses IAM roles to call AWS services (via API Gateway/Lambda or direct API) for content ingestion, user management, billing insights, and analytics dashboards.

3) Core AWS services mapping (recommended roles)
- CDN and delivery
  - CloudFront (distribution with multiple cache behaviors; OAI to restrict S3)
  - CloudFront Functions or Lambda@Edge for request/response customization (auth/paywall gating)
- Identity
  - Amazon Cognito User Pool (authentication)
  - Amazon Cognito Identity Pool (optional temporary credentials)
- Storage and data
  - Amazon S3 (video/assets; versioning; encryption)
  - DynamoDB (Users, Subscriptions, Content catalog, Access logs)
- Video processing and AI
  - AWS Elemental MediaConvert (transcoding)
  - AWS Step Functions (orchestrator)
  - AWS SageMaker (AI/ML inference for captions, recommendations, moderation)
- Compute and admin
  - AWS Lightsail (admin UI/backend)
  - AWS Lambda (serverless glue for API endpoints, small tasks)
  - API Gateway (frontend to Lambda if you prefer a managed API)
- IaC, monitoring, and billing
  - AWS CloudFormation (infrastructure as code)
  - CloudWatch (logs, metrics, dashboards, alarms)
  - CloudTrail (audit logs)
  - AWS Budgets / Cost Explorer (billing management)
- Security
  - IAM roles and policies; KMS encryption keys; WAF on CloudFront; optional Shield

4) Security design patterns
- Access control
  - Use Cognito JWTs to gate content; edge function validates tokens and checks subscription in DynamoDB before serving content.
  - Enable signed URLs/cookies for paid content; rotate keys regularly.

- Data protection
  - Encrypt S3 buckets at rest and in transit; use SSE-KMS with a dedicated key per environment.
  - Encrypt DynamoDB data at rest; enable point-in-time recovery if needed.
- Network
  - Prefer private subnets for internal services; expose only necessary endpoints publicly (via API Gateway/Lambda).
  - Use VPC endpoints for private access to AWS services if needed.
- Compliance and governance
  - Enable AWS Config for resource compliance, and implement proper tagging and cost controls.

5) AI integration touchpoints
- Video processing
  - Transcoding with MediaConvert, with hooks to trigger SageMaker processing (captions, topics, auto-tagging).
- Recommendations and analytics
  - SageMaker endpoints or batch inference to generate personalized content suggestions; store results in DynamoDB.
- Edge optimizations
  - Lightweight edge logic to adjust caching or routing based on simple ML-derived signals (e.g., region-based popularity) without heavy inference at edge.

6) Lightsail integration options
- Admin portal
  - Lightsail-hosted web app for content creators and admins; uses API Gateway/Lambda or an API exposed by Lightsail to manage content, users, and billing.
- Database or microservice (optional)
  - If you need a separate lightweight backend, Lightsail containers can be a simple, predictable option.

7) CloudFormation strategy (IaC)
- Organize into modular stacks:
  - Network and core infra: VPC, subnets, security groups, KMS keys, S3 buckets, DynamoDB tables, Cognito resources.
  - CDN and auth: CloudFront distribution, Lambda@Edge/Functions, OAI, WAF config.
  - Media and AI: MediaConvert role, Step Functions state machine, SageMaker endpoints (optional).
  - Admin and app: Lightsail resources, API Gateway/Lambda backends.
  - Observability and billing: CloudWatch dashboards/alarms, CloudTrail, Budgets.
- Use parameters and mappings to support multi-region or multi-environment deployment.
- Promote changes via a CI/CD pipeline (CodeCommit/CodeBuild/CodePipeline) for CloudFormation change sets.

8) Deployment plan (phased)
- Phase 1: Baseline CDN and identity
  - Set up S3 buckets (videos/assets), DynamoDB tables, Cognito User Pool.
  - Create CloudFront distribution with OAI, basic paywall gating via Lambda@Edge or CloudFront Functions.
  - Build a minimal admin UI on Lightsail that can list content and subscriptions.
- Phase 2: Video pipeline
  - Add MediaConvert, Step Functions workflow, and S3 event-driven triggers for transcoding and thumbnail generation.

- Phase 3: Paywall and access control
  - Implement robust paywall logic at edge/API level; ensure signed URLs/cookies flow and subscription checks.
- Phase 4: AI features (optional in stages)
  - Deploy SageMaker endpoints for captions/recommendations; integrate with DynamoDB and CloudWatch metrics.
- Phase 5: Observability and cost controls
  - Add CloudWatch dashboards, alarms, and budgets; implement logging and tracing.
- Phase 6: Hardening and governance
  - Enforce least privilege IAM policies, WAF/shield, encryption, and compliance checks.
- Phase 7: Testing and go-live
  - Load/test CDN, video processing throughput, paywall gating, and failover/drill.

9) Quick questions to tailor this
- What's your expected traffic footprint (requests per second, video throughput per region)?
- Which regions do you plan to serve from, and do you need multi-region replication?
- Do you want a pure serverless backend (API Gateway + Lambda) or keep Lightsail as your main admin portal?
- What's your paywall model (subscription, per-video, metered, trials), and how will billing be handled (in-app purchases, external payments)?
- Do you need AI features from day one (captions, recommendations, moderation), or do you want to start with the basics and add AI later?
- Data residency or compliance requirements (GDPR, CCPA, etc.)?

10) Next steps
- If you'd like, I can provide:
  - A detailed CloudFormation blueprint (modular stacks) tailored to your environment.
  - A concrete data model for DynamoDB (tables, keys, indexes).
  - A phased bill of materials (SOW-like) with approx. cost ranges by service/usage.
  - Sample AWS CLI commands to bootstrap the core resources.

Would you like me to draft a CloudFormation-ready architecture outline or start with a concrete starter template plan for Core Infra (Cognito, S3, DynamoDB, CloudFront, Lambda), then layer in the video pipeline and paywall next? If yes, tell me your preferred region(s) and whether you want a serverless-first approach or include Lightsail as the primary